VLSI Devices
Lecture 21

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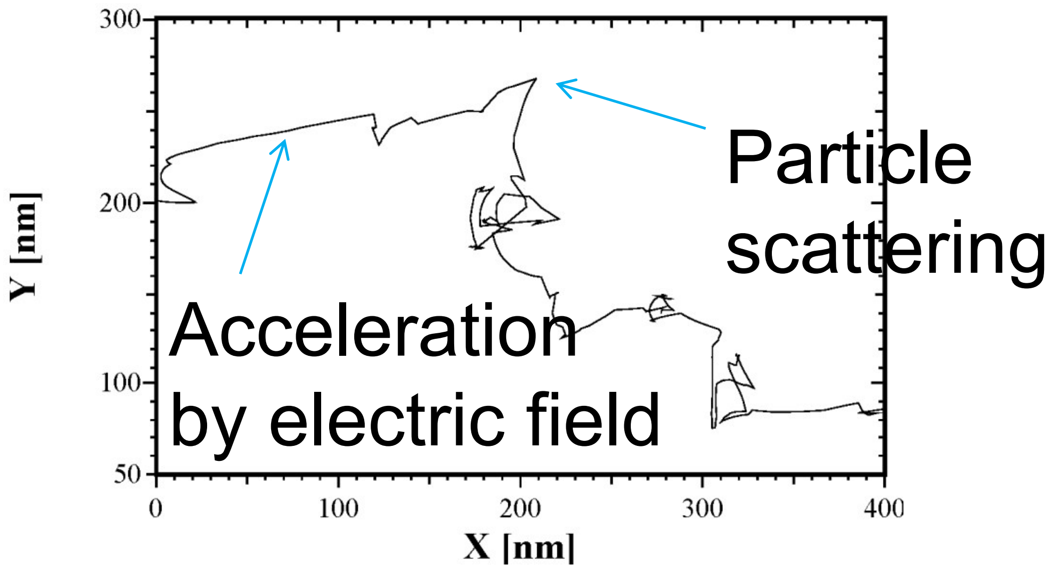
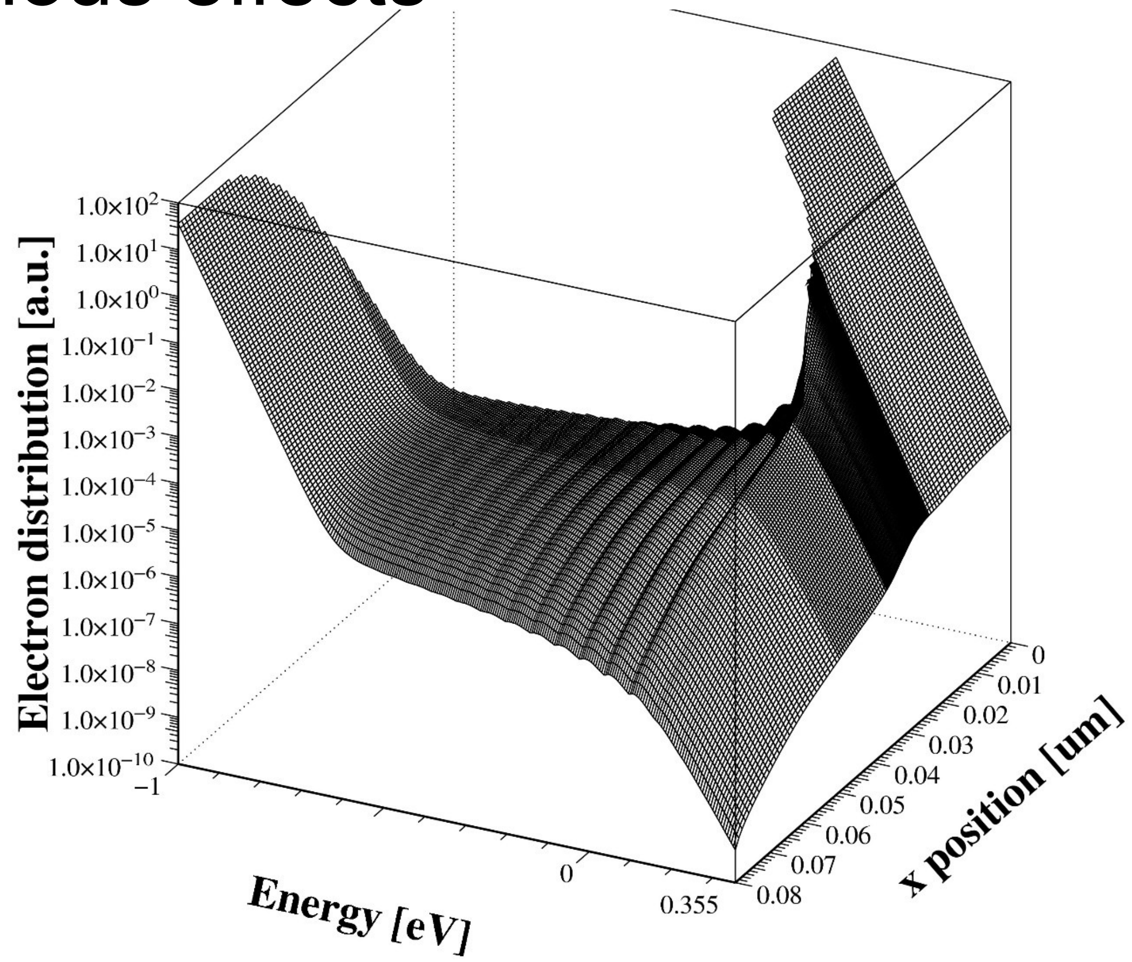
L21

Energy distribution, $f(\mathbf{r}, E)$

- Key quantity to understand various effects
  - Various ways to model it
  - Boltzmann transport equation



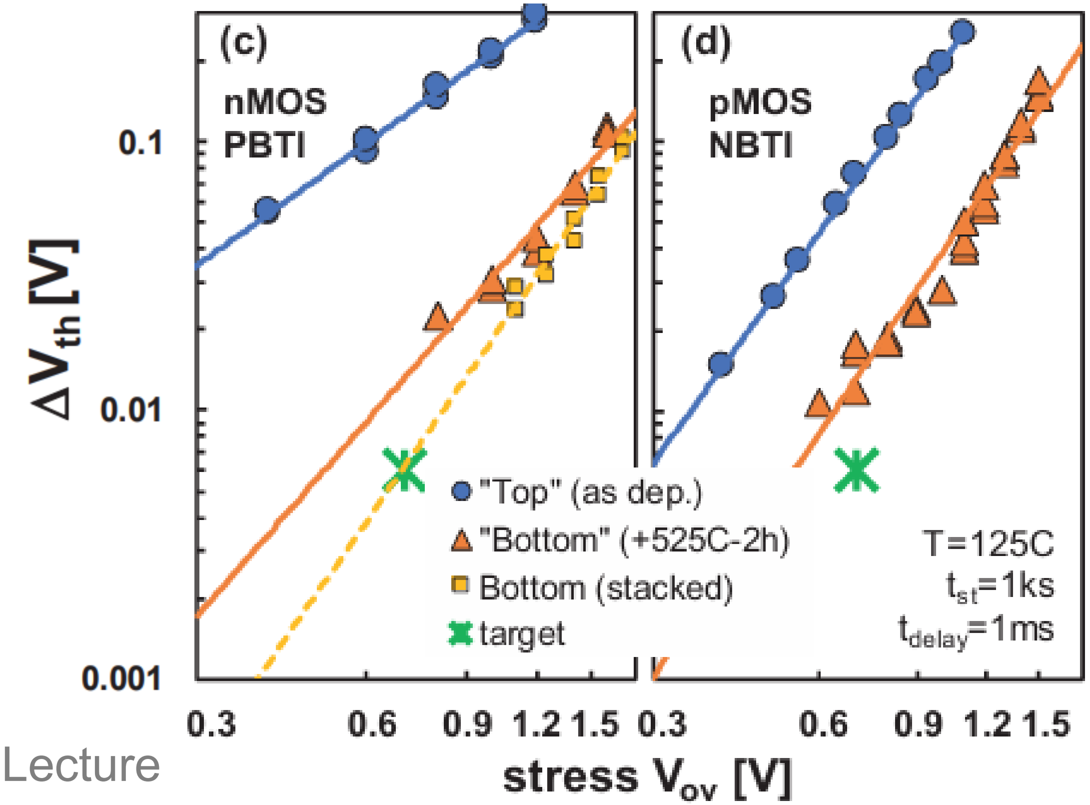
Stochastic electron motion simulated by Monte Carlo



Distribution function

BTI (Bias-Temperature Instability)

- Reliability issue (NBTI in PMOS, PBTI in NMOS)
  - Interface trap generation & $|V_t|$ shift
  - ΔV_{th} < 6mV at V_{ov} = 0.7 V, T = 125 ℃, t_{st} = 1 ks, rescaled from 50mV at 10 years

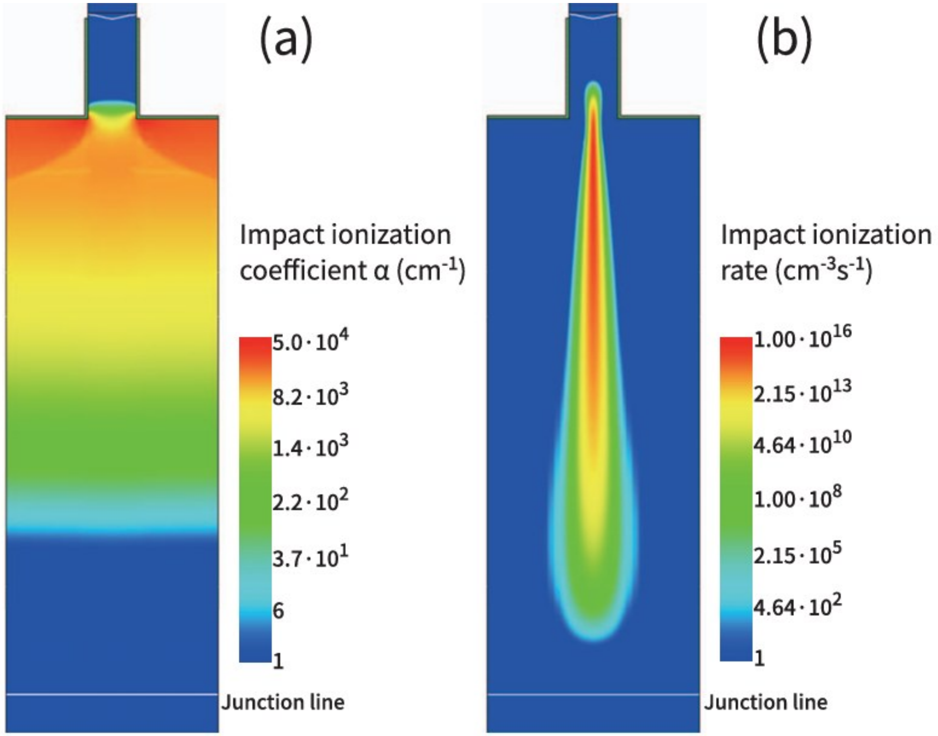


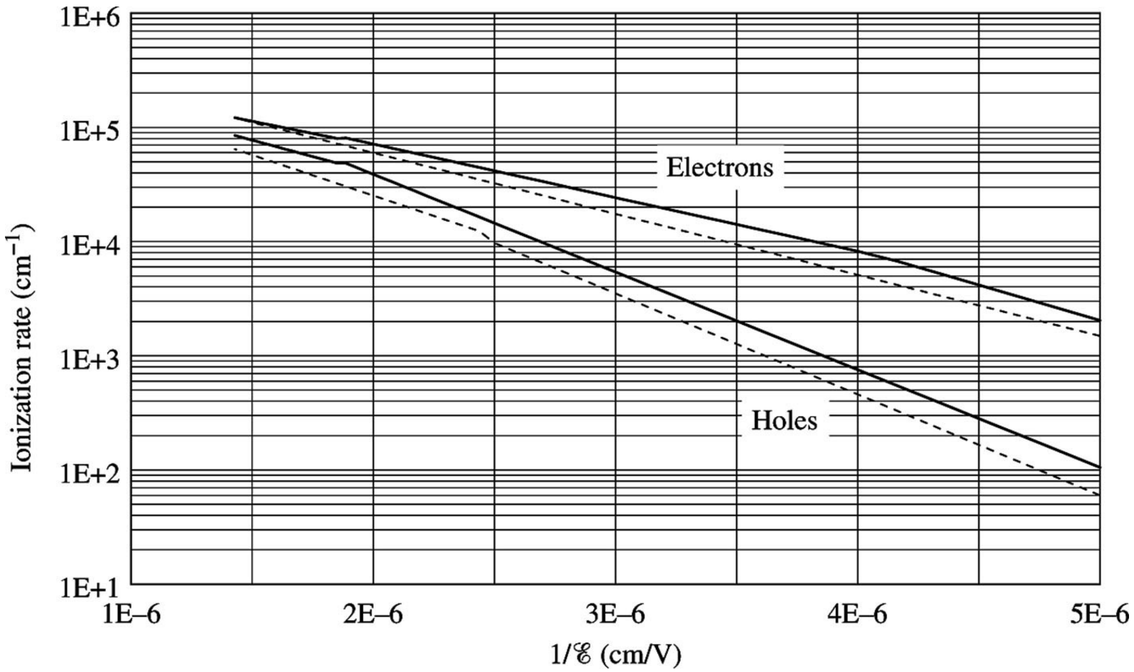
BTI characteristics of gate stacks (IMEC, IEDM 2018)

MOSFET breakdown

- Impact ionization
  - Strong dependence on $\mathcal{E}$

$$\alpha = A\exp\left(-\frac{b}{\mathcal{E}}\right)$$ Taur, Eq. (3.140)

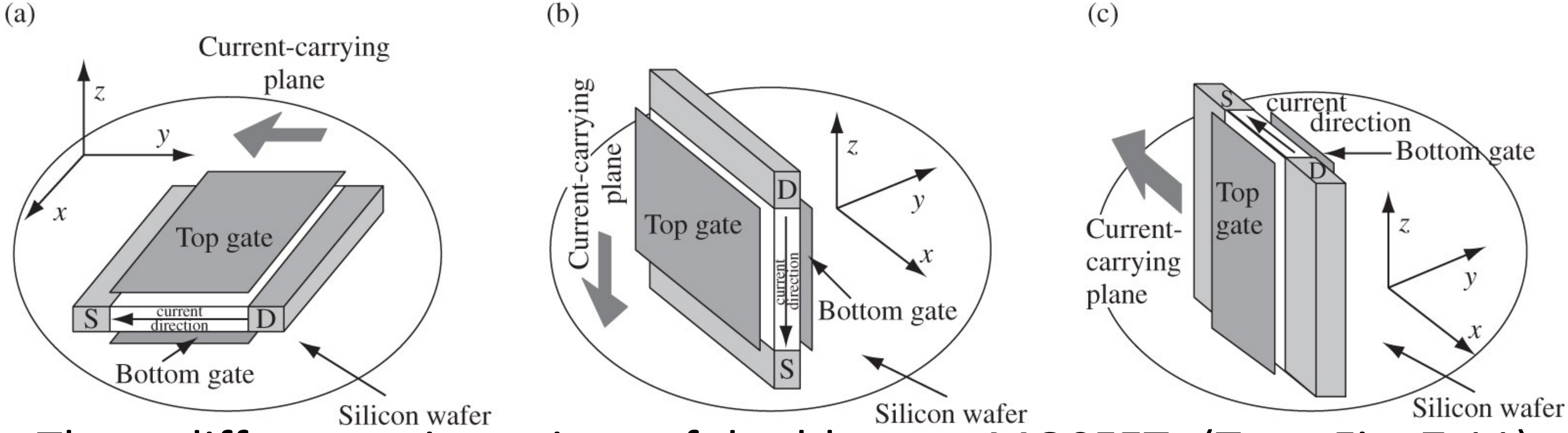




Impact ionization rates in silicon (Taur, Fig. 3.23)

Double-gate (DG) MOSFETs

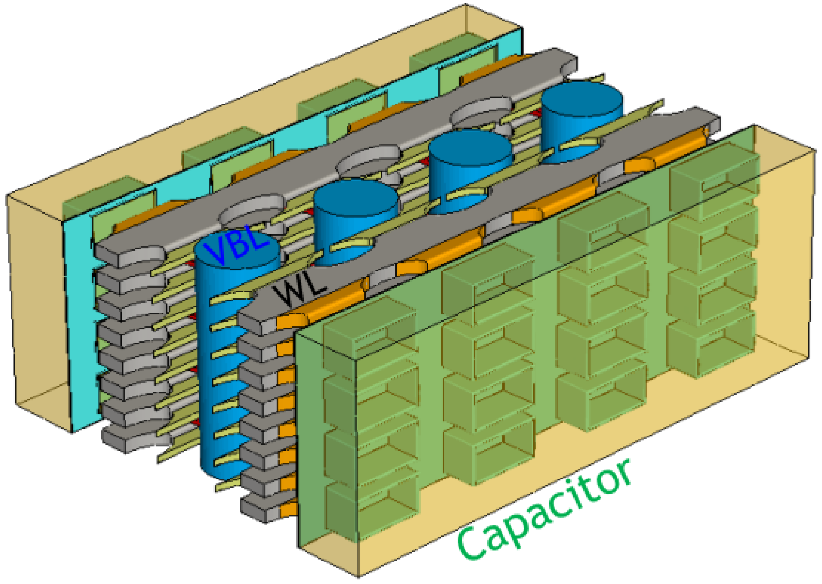
- Advantages of DG MOSFETs
  - Ideal 60 mV/dec subthreshold swing
  - Shorter channel length allowed
  - Channel doping is very low.



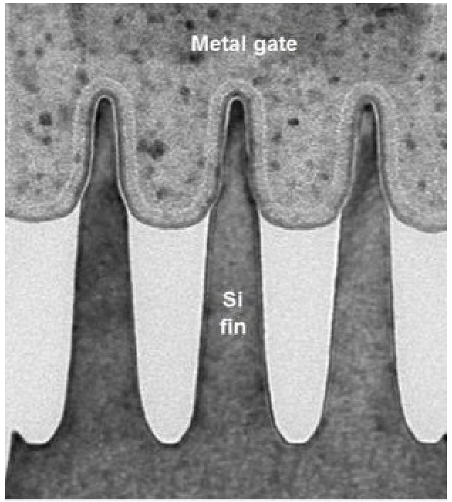
Three different orientations of double-gate MOSFETs (Taur, Fig. 7.11)

Multi-gate MOSFETs

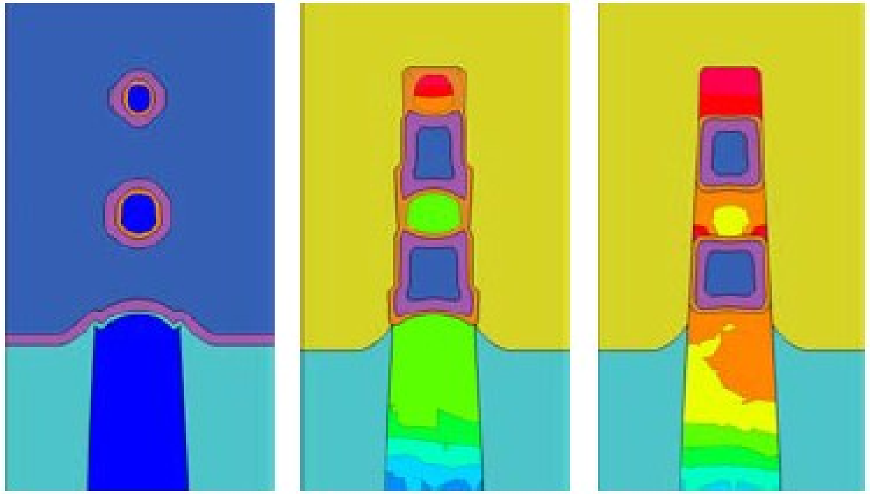
- State-of-the-art transistors
  - Logic: FinFET, Gate-All-Around
  - DRAM: 3D DRAM
  - NAND: Vertical NAND



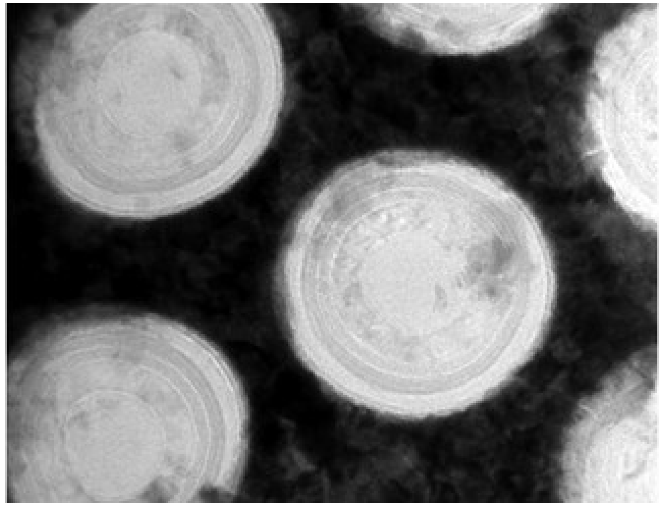
3D DRAM (VLSI 2024)



TEM images of FinFET (Chipworks)



Process-simulated cross-section of stacked NW (IEDM 2016)

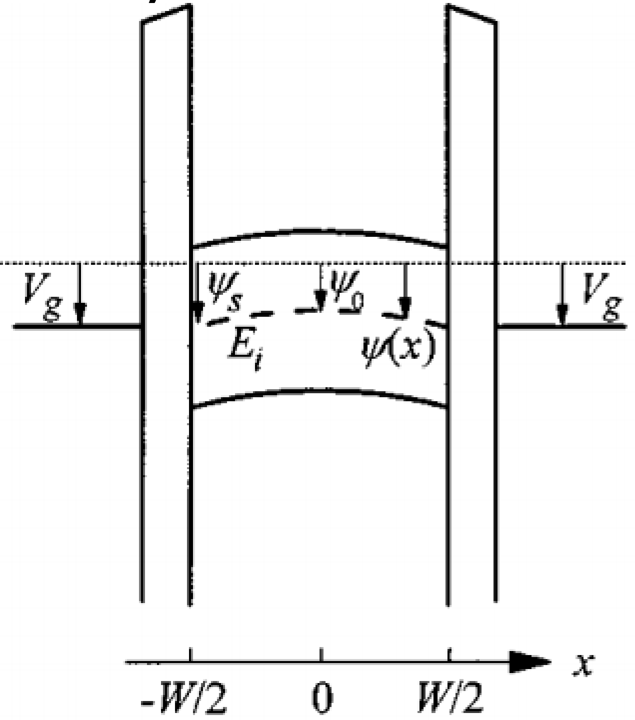


TEM images of vertical NAND (Techinsights)

Main difficulty

- Complicated geometry
  - A planar MOSFET (or a double-gate MOSFET) can be treated as a one-dimensional problem.
  - However, multigate MOSFETs?

Cross section of a nanosheet MOSFET channel



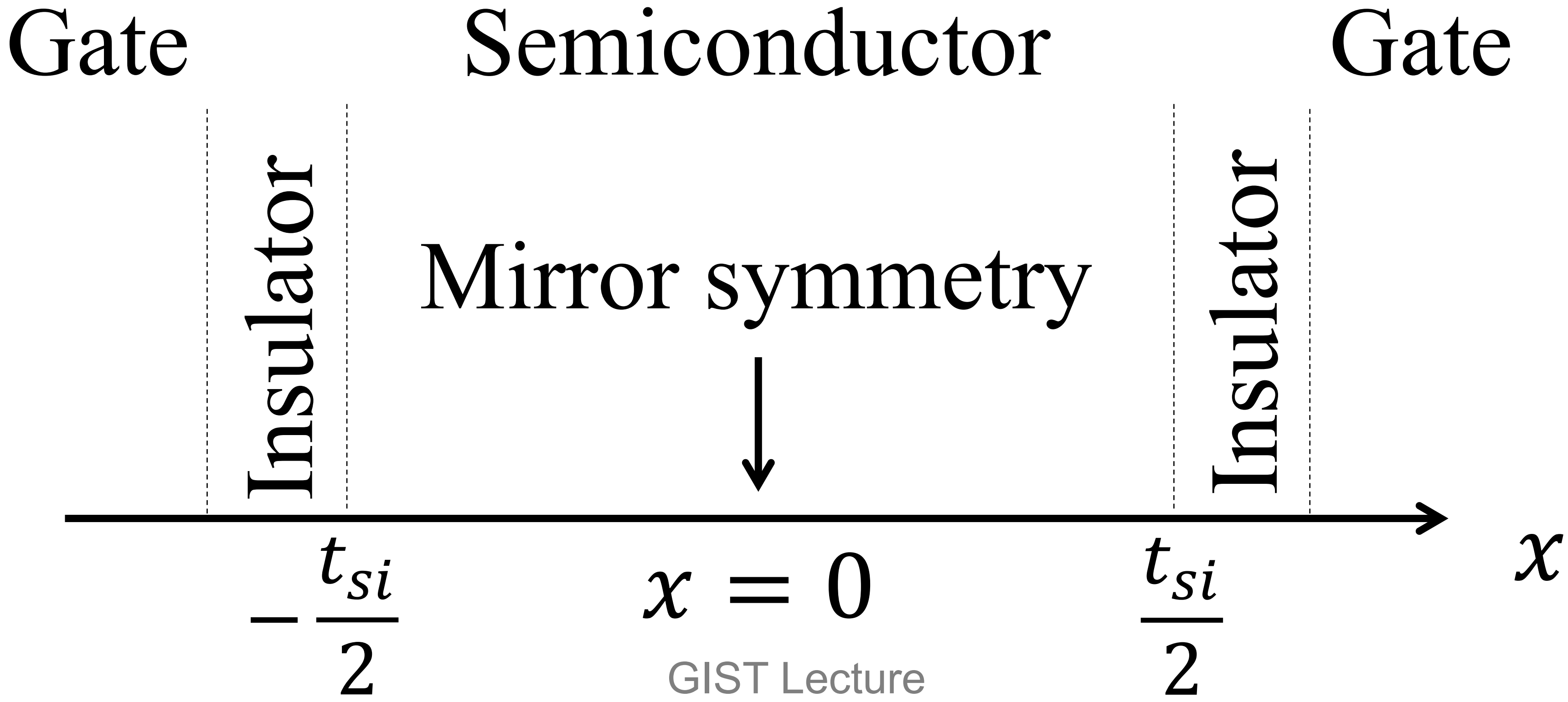
Double-gate MOSFET as a 1D structure (Y. Taur, IEEE EDL, 2000)

Undoped double-gate MOSFET

- Only the mobilie charge

$$\frac{d^2\phi}{dx^2} = \frac{q}{\epsilon_{si}} n_i \exp\left(\frac{q(\phi - V)}{k_B T}\right)$$

Taur, Eq. (7.19)



Integrate it.

- We know the trick. A weighting factor of $\frac{d\phi}{dx}$.

$$\frac{1}{2}\frac{d}{dx}\left(\frac{d\phi}{dx}\right)^2 = \frac{q}{\epsilon_{si}} n_i \exp\left(-\frac{qV}{k_B T}\right) \exp\left(\frac{q\phi}{k_B T}\right)\frac{d\phi}{dx}$$

  - Integration from 0 to x:

$$\frac{1}{2}\left(\frac{d\phi}{dx}\right)^2 = \frac{q}{\epsilon_{si}} n_i \exp\left(-\frac{qV}{k_B T}\right) \int_{\phi(0)}^{\phi(x)} \exp\left(\frac{q\phi}{k_B T}\right) d\phi$$

  - As a result,

$$\frac{1}{2}\left(\frac{d\phi}{dx}\right)^2 = \frac{q}{\epsilon_{si}} n_i \exp\left(-\frac{qV}{k_B T}\right) \frac{k_B T}{q}\left[\exp\left(\frac{q\phi(x)}{k_B T}\right) - \exp\left(\frac{q\phi(0)}{k_B T}\right)\right]$$

Separation of variables

- We assume that $\frac{d\phi}{dx}$ is positive.
  - Then,

$$\frac{d\phi}{\sqrt{\exp\left(\frac{q\phi(x)}{k_B T}\right) - \exp\left(\frac{q\phi(0)}{k_B T}\right)}} = \sqrt{\frac{2k_B T}{\epsilon_{si}} n_i \exp\left(-\frac{qV}{k_B T}\right)} dx$$

  - With normalized potential variables,

$$\frac{d\tilde{\phi}}{\sqrt{\exp\left(\tilde{\phi}(x)\right) - \exp\left(\tilde{\phi}(0)\right)}} = \sqrt{\frac{2q^2 n_i}{\epsilon_{si} k_B T} \exp(-\tilde{V})} dx$$

Manipulation

- LHS must be integrated.

$$\frac{1}{\sqrt{\exp\left(\tilde{\phi}(0)\right)}}\frac{d\tilde{\phi}}{\sqrt{\exp\left(\tilde{\phi}(x)-\tilde{\phi}(0)\right)-1}}$$

  – With a new variable, $u \equiv \tilde{\phi}(x) - \tilde{\phi}(0)$, ($u$ is positive.)

$$\frac{1}{\exp\left(\frac{\tilde{\phi}(0)}{2}\right)}\frac{du}{\sqrt{\exp(u)-1}}$$

  – Then, the main task is

$$\int \frac{du}{\sqrt{\exp(u)-1}}$$

Introduce $v^2 = \exp(u) - 1$.

- It follows $2vdv = \exp(u)\,du$.
  - The integration becomes
$$\int \frac{du}{\sqrt{\exp(u) - 1}} = \int \frac{\exp(-u)2vdv}{v} = 2\int \frac{dv}{v^2 + 1}$$
  - It is a known integral.
$$\int \frac{dv}{v^2 + 1} = \arctan v + C$$
  - In terms of u,
$$\int \frac{du}{\sqrt{\exp(u) - 1}} = 2\arctan\left(\sqrt{\exp(u) - 1}\right) + C$$

Entire LHS and RHS terms

- When u varies from 0 to $\tilde{\phi}(x) - \tilde{\phi}(0)$,
  - Then, the entire LHS term becomes

$$\frac{2}{\exp\left(\frac{\tilde{\phi}(0)}{2}\right)} \arctan\left(\sqrt{\exp\left(\tilde{\phi}(x) - \tilde{\phi}(0)\right) - 1}\right)$$

  - Of course, the RHS term is

$$x\sqrt{\frac{2q^2 n_i}{\epsilon_{si} k_B T} \exp\left(-\tilde{V}\right)}$$

Matching them

- By taking tangent,

$$\sqrt{\exp\left(\tilde{\phi}(x)-\tilde{\phi}(0)\right)-1}=\tan\left[x\sqrt{\frac{q^2 n_i}{2\epsilon_{si}k_B T}\exp\left(\tilde{\phi}(0)-\tilde{V}\right)}\right]$$

  – A simple manipulation yields

$$\exp\left(\tilde{\phi}(x)-\tilde{\phi}(0)\right)=\sec^2\left[x\sqrt{\frac{q^2 n_i}{2\epsilon_{si}k_B T}\exp\left(\tilde{\phi}(0)-\tilde{V}\right)}\right]$$

  – Now we have

$$\tilde{\phi}(x)-\tilde{\phi}(0)=-2\log\left[\cos\left(x\sqrt{\frac{q^2 n_i}{2\epsilon_{si}k_B T}\exp\left(\tilde{\phi}(0)-\tilde{V}\right)}\right)\right]$$

Thank you!